GG-B-140D
10 November 1986
SUPERSEDING
GG-B-00140C (DSA-DM)
17 September 1970 and
GG-B-140A
16 April 1957

FEDERAL SPECIFICATION

BATH, WATER, DENTAL IMPRESSION COMPOUND

This specification is approved by the Commissioner Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an electrically-heated, thermostatically-controlled water bath and insert pan suitable for dental use.

1.2 Classification. The complete water bath and insert pan replacement part shall be furnished in the type specified below (see 6.1).

Type I. Complete water bath.

Type II. Replacement insert pan.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Federal Specifications

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Federal Standards

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of listed federal and military standards, specifications, and Commercial Item Descriptions (CIDS) listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.

Military Specifications

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or requests for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI C73.11	ANSI Standard for Outlet Receptacles, Attachment Plug Caps and Appliance Plugs.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

UNDERWRITERS' LABORATORIES, INC.

Underwriters' Laboratories Standard for Industrial Control Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., NY 11746; 207 E. Ohio Street, Chicago, IL 60611; 1655 Scott Blvd., Santa Clara, Ca 95050;333 Pfingsten Road, Northbrook, IL 60062.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Prior to delivery the contractor shall submit proof to the contracting agency that the equipment he will supply under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. Standards. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

3.1.2 In lieu of the label or listing, the contractor may submit independent proof satisfactory to the contracting agency, prior to delivery, that his equipment conforms to the Underwriters' Laboratories, Inc. Standards.

3.1.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of this commodity.

3.2 Materials. Materials shall conform to the applicable requirements specified herein. Materials not specifically designated shall be of high commercial quality throughout and shall be entirely suitable for the intended purpose.

3.2.1 Steel. Sheet steel for the enclosure, removable insert pan, cover, shell, and lifting platform shall comply with the chemical composition for AISI type 302 or 304 steel of FED-STD-66.

3.2.2 Rubber. Rubber for the perforated rack shall be black, Government Rubber-Styrene (GR-S) synthetic rubber with a Durometer hardness of B-75 to B-85.

3.3 Design. Type I shall be a complete water bath and type II shall be a replacement insert pan for the water bath. The configuration and arrangement of the complete water bath shall conform to figures 1 and 2. The water bath shall be suitable for operation on 115 volt, 50/60 hertz, alternating current. The design of the insert pan, both when furnished as part of type I and a separate replacement part (type II), shall conform to figure 4. In figure 4, the section A-A dimension 1/32 inch, plus 1/32, minus 0, shall refer to the depth of rib and not to the gauge of the metal. The unit shall be capable of heating water to a thermostatically controlled temperature ranging from 90 degrees Fahrenheit (^{O}F) to boiling. The unit shall not damage any surface by heat or friction, on which it might be placed in normal operation. The configuration and fit of all components shall eliminate any cracks and crevices in which materials might collect, and the unit in general shall be readily cleanable by conventional means. All components of the unit shall be corrosion-resisting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

3.3.1 Figures. The figures forming a part of this specification are specification control figures. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within those prescribed on the drawings to insure proper design, function, fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

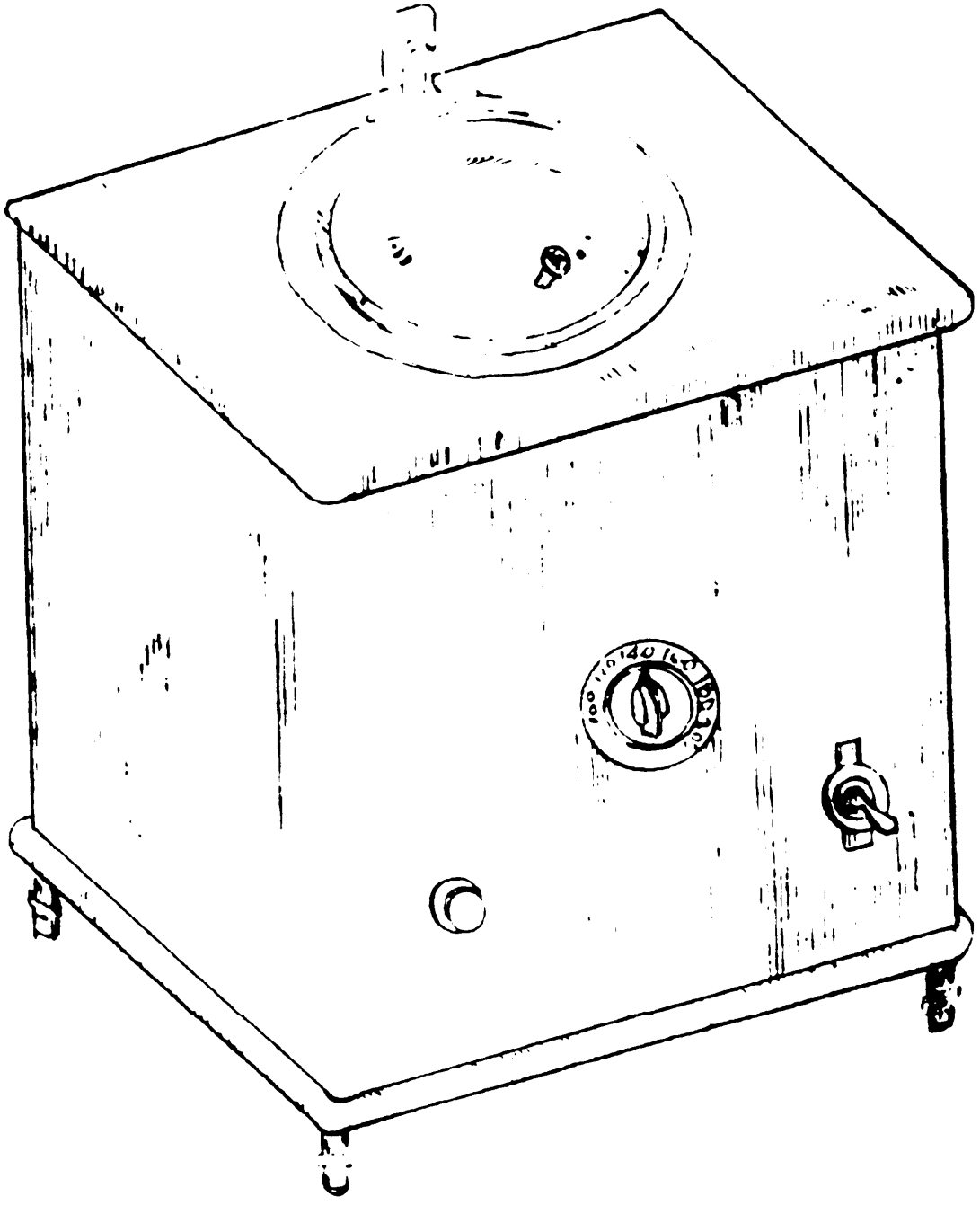
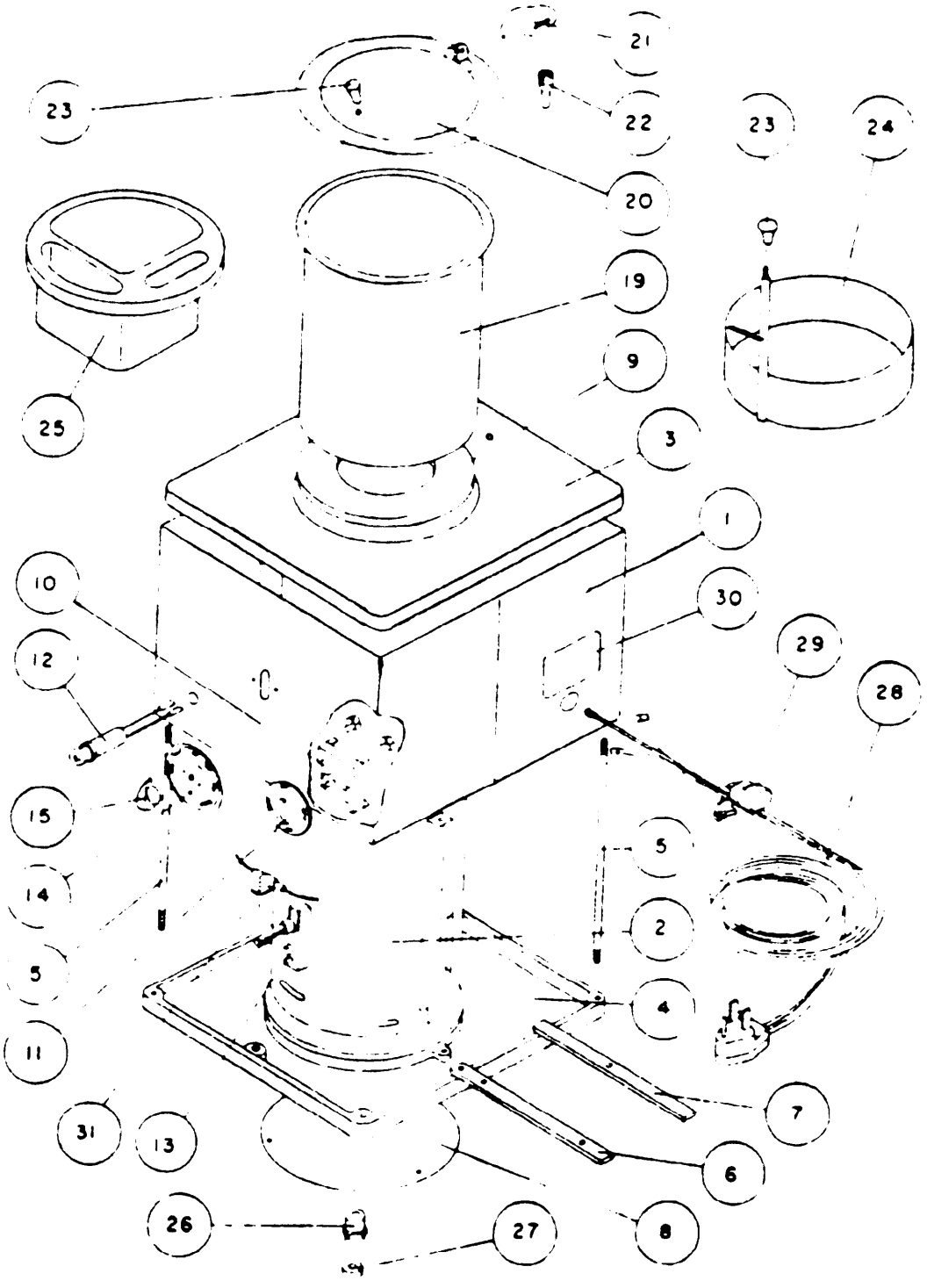


FIGURE 1

ISOMETRIC VIEW

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.



EXPLODED VIEW

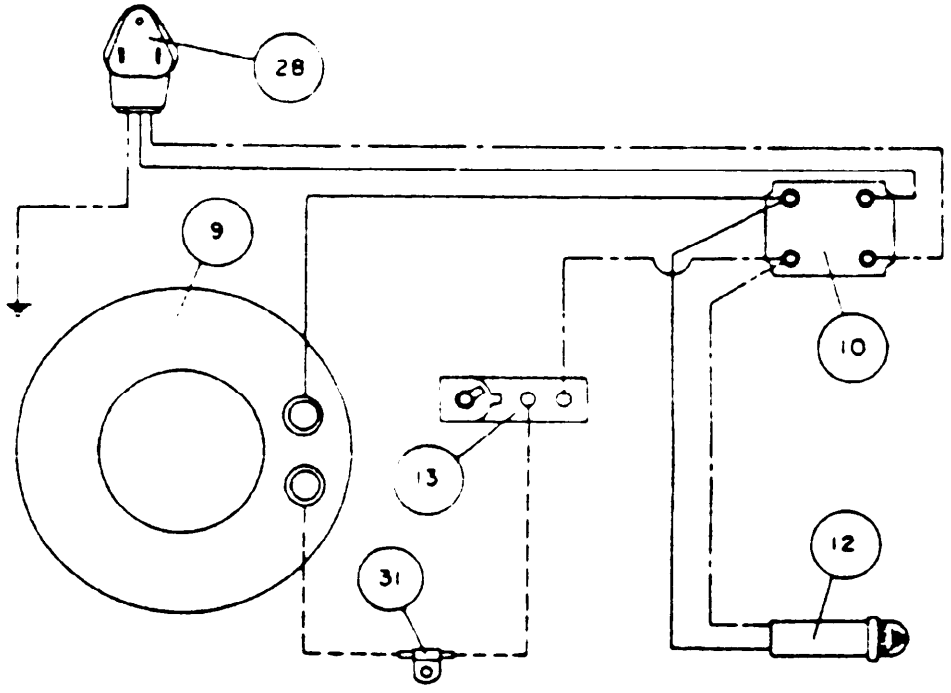
FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

SUBSTITUTION TO THE EXTENT SPECIFIED HEREIN.

PARTS LIST (See Figures 2 and 3)

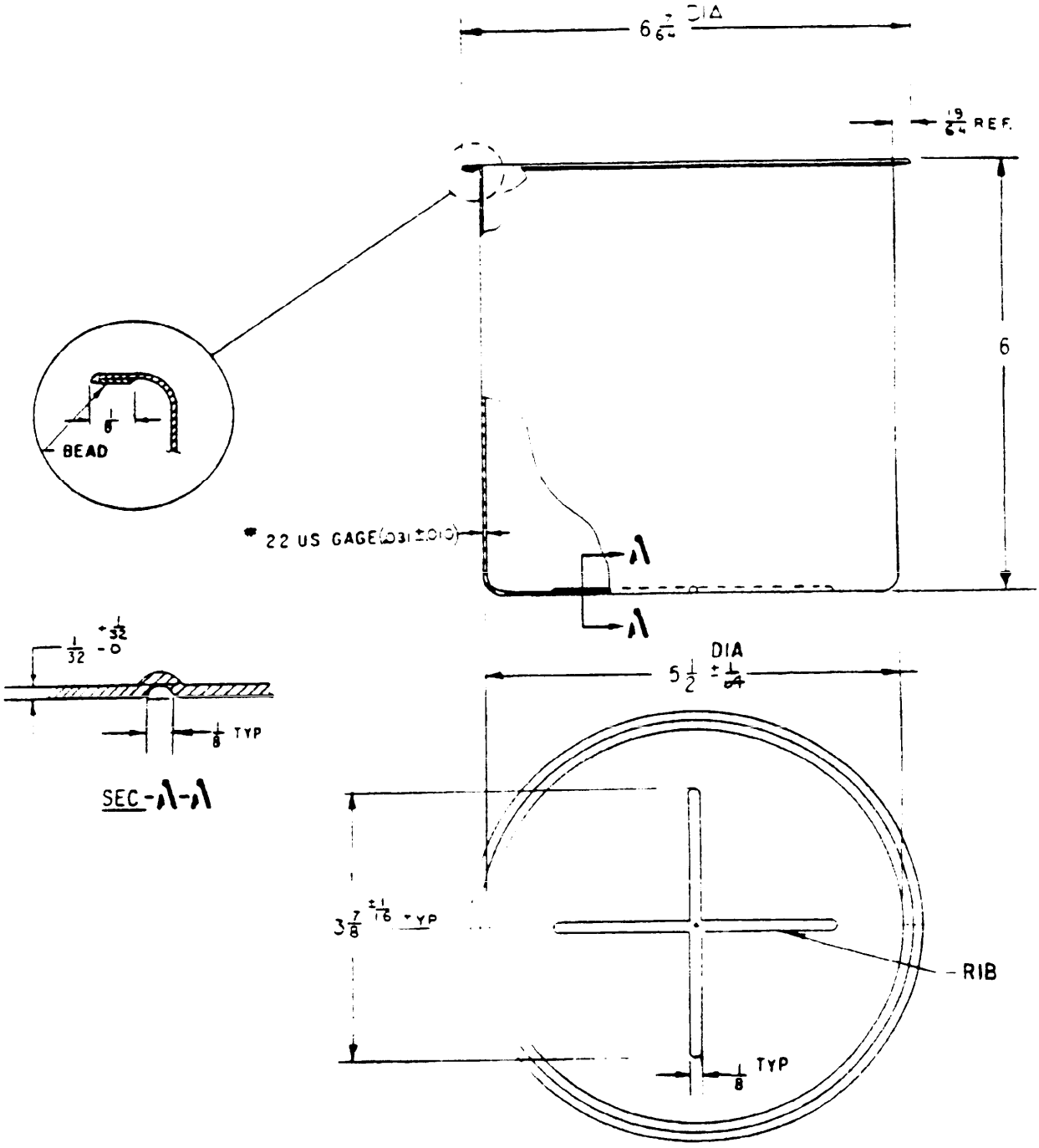
PART NO.	PART NAME	NO. REQ'D	PART NO.	PART NAME	NO. REQ'D
1	Enclosure	1	15	Pointer Knob, Plastic	1
2	Tubular Shell (Inner Shell)	1	19	Insert Pan	1
3	Top, Die Casting	1	20	Cover	1
4	Bottom, Die Casting	1	21	Hinge	1
5	Corner Stud	4	22	Post for Hinge	1
6	Element Support (2 rivets)	1	23	Knob, Plastic	2
7	Element Support (1 rivet)	1	24	Lifting Platform	1
8	Heat Shield	1	25	Perforated Rack	1
9	Heating Element	1	26	Leg, Black, Anodized	4
10	Main Switch	1	27	Rubber Cushion	4
11	Knob for Switch	1	28	Electrical Cord	1
12	Pilot Light	1	29	Strain relief	1
13	Thermostat	1	30	Name Plate	1
14	Temperature Dial	1	31	Thermal Cut-Off	1



WIRING DIAGRAM

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.



NOTE

1. ALL EDGES SHALL BE ROUNDED
2. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES AND TOLERANCES SHALL BE ± 1/32

BOTTOM VIEW

FIGURE 4

INSERT PAN, WATER BATH

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

3.4 Construction.

3.4.1 General. The type I water bath shall consist of a stainless steel enclosure and an inner shell, held between two die cast aluminum plates which form the top and bottom. This assembly shall be held together by a vertical aluminum stud at each of the four corners of the unit. Each stud shall screw into the underside of the top and pass through the bottom, where a black anodized aluminum leg and a black rubber cushion shall be secured to it. The outside dimensions of the unit shall not exceed 9-1/2 inches in length, 8-1/4 inches in width and 9-1/2 inches in height when the cover is in place. The unit shall have several integral components, as specified in 3.5.1, and removable components as specified in 3.5.2. Hardware, such as screws, nuts, and hinges, shall be stainless steel or other suitable metal treated to render it corrosion-resisting.

3.4.2 Enclosure. The square-shaped side enclosure shall be fabricated of no less than 0.025 inch (No. 24, United States Standard Gage) stainless steel, as specified in 3.2.1. Assembled, it shall form the four sides of the water bath. The side enclosure shall leave an insulating dead air space of one inch or more surrounding the tubular shell.

3.4.3 Tubular shell. The tubular shell shall be fabricated of no less than 0.025 inch (No 24. United States Standard Gage) stainless steel, as specified in 3.2.1, closely fitting the insert pan, and assembled into circular openings of the die-cast aluminum top and bottom. The shell shall have a thermostat mounted on the vertical wall so that it touches the insert pan at the lower end of its vertical height. Two pairs of slots shall be provided on the circumference of the shell, with a pair at 120 circular degrees in each direction from the thermostat. The slots shall approximate the length of the thermostat and shall be positioned on the same level as the thermostat. The shell material between the slots shall be deformed to form two springs which shall exert sufficient force to hold the tank against the thermostat.

3.4.4. End plates. The top and bottom shall be of sufficient thickness to permit drilling and tapping where required and, together with the studs and related hardware, shall provide for the general stability of the unit.

3.5 Components.

3.5.1 Integral components.

3.5.1.1 Element support. Two element supports shall provide stable support for the heating element during normal operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

3.5.1.2 Heat shield. A heat shield shall be provided which will limit the temperature of the surface under the unit to 150 degrees Fahrenheit (°F.), when tested as specified in 4.5.2.

3.5.1.3 Electrical heating system. The system shall consist of the components specified below, located as shown in figures 1 and 2, wired as shown in figure 3, and functioning as required in 3.3. The system shall be capable of reaching the specified power level of 300 watts plus or minus 10 percent, when tested as specified in 4.5.4.2. The system shall exhibit no signs of electrical, thermal, or mechanical deterioration during or immediately after the cycling test specified in 4.5.4.1. The system shall be capable of withstanding without failure a dielectric test as specified in 4.5.4.3.

3.5.1.3.1 Heating element. The heating element shall be of the disk or sheathed ring type. The wire shall be of nickel-chromium alloy, imbedded in compressed refractory material and enclosed in a monel metal sheath. It shall have heat-resisting terminals, shall be connected by nickel wire with a heat-resisting jacket, and shall be rated to provide a uniform temperature. In addition to the tests specified for the entire electrical heating system, the heating element shall be capable of withstanding the continuous duty test specified in 4.5.3.

3.5.1.3.2 Main switch. The main switch shall be of a 15 ampere rating at 125 volts. It shall be a double pole reciprocating type with a rotating knob dial plainly marked "ON-OFF-ON-OFF". The main switch shall serve only as a total disconnecting switch.

3.5.1.3.3 Pilot light. The pilot light shall be the neon type, with a nylon body and a red plastic lens. It shall indicate only when the unit is in operation.

3.5.1.3.4 Thermostat. The thermostat shall be of strut-type construction, enclosed in a stainless steel case and located on the front side of the unit. It shall contact the insert pan at all times when the insert pan is telescoped into the shell. The thermostat shall be easily removable for service without the necessity of disassembling the entire unit. The thermostat shall maintain the set temperature such that the temperature reading (after stabilizing) shall not vary more than 2°F. and the range of temperature at any given setting shall fall within 5°F. of the temperature setting, when tested as specified in 4.5.4.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

3.5.1.3.5 Temperature dial. The temperature dial shall read from 90°F. to boiling. The calibrated temperature of the thermostat shall be permanently marked on the dial in 20-degree increments, starting at 100 degrees. A plastic pointer-knob shall be furnished to indicate the temperature on the dial. A reference line for one extreme position of the thermostat shall be provided on the dial to facilitate proper location of the pointer knob (assembled to the thermostat) in relation to the temperature dial. The dial calibration shall be accurate to plus or minus 5 degrees Fahrenheit over the range of 90°F to boiling, when tested as specified in 4.5.4.4.

3.5.1.3.6 Electrical cord. Each unit shall be provided with an electric cord, no less than 6 feet in usable length. The cord shall be 3-wire, No. 16 or 18 gage, type SPT-3 thermosplastic insulated or type HPN, HSJ or HSJO rubber jacketed cord. A 3-prong vinyl plug shall be attached to one end and shall be suitable for use with receptacles specified in American National Standard Institute (ANSI) Standard C73.11 for caps and receptacles; 125 volts, 16 amperes, 2 pole, 3-wire grounding type. The opposite end shall be slit back and stripped for attachment of terminals for direct connection to main switch. The cord shall be provided with a nylon strain relief.

3.5.2 Removable components.

3.5.2.1 Removable insert pan. The removable insert pan, both when furnished as part of type I and as a separate replacement part (type II), shall be formed from one piece of stainless steel and shall be furnished with a rolled, flattened bead on the top edge to provide a finger grip for removal. The bottom of the pan shall be ribbed for reinforcing purposes and for preventing this surface from developing concavity which would affect its heating capability. The pan shall have a telescopic fit into the tubular shell of water bath, shall be readily removable, and shall have full surface to surface contact with the heating element when placed in position. When the insert pan is ordered as a replacement part, the perforated rack, metal cover, and lifting platform shall not be furnished.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

3.5.2.2 Perforated rack. The removable insert pan, when furnished as part of type I only, shall be furnished with a three-compartmented perforated rack. One perforated compartment measuring 3/4 inch wide and 5/8 inch deep shall accommodate tubes of hydrocolloid, the second 5/8 inch deep perforated compartment shall accommodate syringes ready for immediate use, and the third compartment shall be a 2-1/2 inch deep, nonperforated tempering bath with 3/16 inch minimum side wall thickness and 1/2 inch minimum bottom wall thickness, designed to receive the filled impression tray. The rack shall be fabricated of rubber as specified in 3.2.2.

3.5.2.3 Cover. A cover shall be provided for the insert pan (when furnished as part of type I only). The cover for the removable insert pan shall be one-piece, of the same material as the removable insert pan, and shall measure 6-3/8 inches in outside diameter. It shall be dome-shaped with a reverse draw, to provide a drip ledge inside the tank for the condensate to drip back into the tank when the cover is in place over the insert pan. A plastic knob shall be secured to the cover to provide a means for lifting the cover. A hinge shall be secured to the back of the cover and fastened to the top of the unit. The hinge shall permit three positions of the cover: cover in place on the tank, cover in a nearly vertical position to permit condensate to drain into the tank when open, and cover tilted all the way over to permit removal of the pan from the unit.

3.5.2.4 Lifting platform. A lifting platform for convenient submersion and removal of materials shall be provided for the insert pan (when furnished as part of type I only). It shall be made of stainless steel as specified in 3.2.1 and shall be a ring 4-3/4 inches in diameter and 1-1/2 inches high. A vertical handle shall be provided with a plastic knob for lifting out of the tank. A cloth cover to fit over the ring shall be provided on which may be laid the articles to be suspended in the water bath. Two of these cloth covers are to be furnished.

3.6 Tolerances. Unless otherwise specified herein, tolerances on dimensions shall be $\pm$ 5 percent providing fit and function is not sacrificed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the currentversion before use.

3.7 Service data. The contractor shall furnish two copies of a service data manual with each complete water bath (type I). In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. This service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. Such detailed information shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only

3.7.1 Approval of service data. As soon as practicable after award of the contract or purchase order, the contractor shall furnish two preliminary copies of his proposed service data to the contracting officer, via the cognizant Government inspector, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall by submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow the contractor to incorporate required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

3.7.2 Waiver of review and distribution copies of manual. Should the supplier have furnished manuals which have been reviewed, commented on, and approved by the Directorate of Medical Materiel for the identical item within two years preceding the date of contract, and should the supplier propose to furnish manuals identical to those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.8 Finish. After all surface imperfections have been removed by grinding, the inside and outside surfaces of platform, removable insert pan and cover, and exposed surface of inner shell, shall be finished with a No. 4 finish. As an alternate, the interior surface of the insert pan may be electropolished. The finish shall result in a uniform, bright, shiny surface which may be readily cleaned. All surfaces shall be smooth and free from sharp edges. As an alternate, the exposed surface of the inner shell shall be of a No. 2B finish.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the currentversion before use.

3.9 Identification marking. Each water bath shall have a corrosion-resisting pressure sensitive tape nameplate permanently affixed thereto. The bath shall bear the model or style designation, voltage, watts, frequency, and the name or registered trademark of the manufacturer. The markings on the plate shall be permanently and legibly printed and a serial number shall be permanently depressed in the plate by the use of routine typewriter characters. Type II, replacement insert pan shall have the name or registered trademark of the manufacturer die-stamped on the face of the pan. As an alternate, this information shall be permanently etched in one of the quadrants formed by the rib on the bottom, outer surface of the pan.

3.10 Workmanship. Water baths and insert pans shall be free from defects which detract from their appearance or may impair their serviceability.

3.11 Performance test. Each water bath shall be given performance test as specified in 4.5.4.5 (see 4.4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, rubber, and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Classification of inspection. The inspection of the water bath (type I), and insert pan (type II), shall be classified as follows:

1. First article inspection.
2. Quality conformance inspection.

4.3 First article inspection. When a first article is specified in the contract, the first article inspection shall consist of all examinations specifed in 4.4.2 and all testing specified in 4.5 (as applicable for the specific type), and shall apply to the first article sample only (see 6.1).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of all examinations specified in 4.4.2, and the testing specified in 4.5.1, 4.5.4, 4.5.4.3, 4.5.4.4 and 4.5.4.5 (as applicable for the specific type), and shall apply to those samples selected in accordance with 4.4.1 and 4.4.3. Quality conformance inspection shall be performed by the supplier in accordance with 4.1.

4.4.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table I. Unit of product for examination purposes shall be one complete water bath (type I) or one insert pan (type II).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the current version before use.

Table I. Sampling and Acceptance Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
For visual examination		
Major	II	1.0
Minor	II	2.5
For dimensional examination	S-2	2.5

4.4.2 Examination. The water bath and insert pan shall be examined to determine compliance with all requirements contained in the specification, and shall be examined for defects, including, but not limited to, those listed in table II.

Table II. Classification of Defects

Categories	Defects
Major:	
101	Water bath not wired as specified.
102	Fit between insert pan and shell loose.
103	Insert pan cover construction not as specified.
104	Surface imperfections of pan not removed by grinding.
105	Ribs not present in bottom of pan.
106	Finger grip on pan not as specified.
107	Gage of pan metal not as specified.
Minor:	
201	Finish not as specified.
202	Legs not as specified.
203	Control location not as specified.
204	Identification marking missing or illegible.

4.4.2.1 Dimensional examination. The water bath and insert pan shall be examined for defects in dimensions. Any dimension not within the tolerance specified in 3.6 shall be classified as a defect.

4.4.3 Sampling for testing. Sampling for testing shall be in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. The unit of product shall be one complete water bath (type I) for tests specified in table III.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the currentversion before use.

Table III. Sampling and Acceptance Criteria for Testing

Characteristic	Requirement	Test Procedure	Inspection Level
Heat shield effectiveness	3.5.1.2	4.5.2	S-2
Continuous duty	3.5.1.3.1	4.5.3	S-2
Cycling	3.5.1.3	4.5.4.1	S-2
Power level	3.5.1.3	4.5.4.2	S-2
Dielectric strength	3.5.1.3	4.5.4.3	100%
Calibration	3.5.1.3.5	4.5.4.4	S-2
Performance test	3.11	4.5.4.5	100%*

*Performance test shall be considered performed on those water baths that were subject to quality conformance testing.

4.4.4. Sampling for raw material testing. Sufficient amount of each lot of each metal used in the construction of the stainless steel components of the water bath and for the insert pan when furnished as either part of type I, or as type II shall be selected for chemical analysis as required by 3.2.1 and specified in 4.5.1.

4.4.5 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements. Tests specified in 4.5.2 through 4.5.4.4 shall be conducted only on type I water bath.

4.5.1 Chemical analysis. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

4.5.2 Heat shield effectiveness. The water bath shall be operated at the boiling setting, with the insert pan filled, until the temperature at the center of the heat shield surface under the unit reaches maximum. This test shall be conducted on a black nonreflective surface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

4.5.3 Continuous duty. During the first article test procedure, the heating element shall be operated continuously for 23 hours at a power level of 300 watts ± 10 percent and at an operating temperature of 200°F., followed by one hour of operation with the thermostat set for boiling. Performance shall be consistent and shall show no signs of deterioration of the element. The test period for quality conformance inspection shall be for 7 hours at the 200°F temperature setting and one hour with the thermostat set for boiling.

4.5.4 Operational tests. Operational tests shall be performed on the water bath to obtain the test data required for the cycling, power level, and calibration tests specified below. Preparation for test shall consist of immersing an accurate thermometer in center of the insert pan filled with water to proper operating level and connecting a voltmeter and ammeter in the heating element branch of the circuit. Height above sea level shall be considered in determining the thermostat setting necessary in the upper temperature range to produce a water boiling condition. First article test procedure shall consist of recording temperature dial, thermometer, voltmeter, and ammeter readings during the test period. Test values shall be set on the thermostat at increments of 10°F. between 90°F. and boiling. Quality conformance test procedure shall parallel the first article procedure except the thermostat test increments shall be 20°F. over the 90°F. to boiling range.

4.5.4.1 Cycling. Cycling test of the electrical heating system shall be performed by checking the open and close operation of the thermostat during each test interval of the operational test procedures. The operation of the theromstat shall be checked to insure that the water temperature is being raised in conformance with test requirements and that a water boiling condition is eventually provided by the water bath components. Only heating elements which have successfully passed the continuous duty test specified in 4.5.3 shall be used as components for this test.

4.5.4.2 Power level. The product of the voltmeter and ammeter readings recorded during each test interval of the operational test procedures shall be used to determine the heating systems capability of reaching the power level specified in 3.5.1.3. Test results shall be uniform throughout and shall not exceed the tolerance specified. Only water baths which have successfully passed the cycling test specified in 4.5.4.1 shall be used for this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

4.5.4.3 Dielectric strength test. All electrical component parts shall be tested after complete assembly and shall withstand without failure, a dielectric test of twice the rated voltage plus 1,000 volts or a total of 1,250 volts applied for one minute. This test shall be made on each of the individual lines of the circuit, using the metal frame as the other leg or ground of the 1,250 volt test. Only water baths which have successfully passed the power level test specified in 4.5.4.2 shall be used for this test under first article inspection.

4.5.4.4 Calibration test. The thermostat and thermometer readings, recorded during each interval of the operational test procedures, shall be used to check the heating system accuracy of calibration and thermostat control. At each point, the temperature reading and temperature range shall meet the requirements of 3.5.1.3.4. At each point, the temperature reading (after stabilizing) shall not vary more than 2°F. and the range of temperature at any given setting must fall within 5°F. of the temperature setting. Only water baths which have passed the dielectric strength test specified in 4.5.4.3 shall be used for this test.

4.5.4.5 Performance test. Each water bath shall be given a performance test in accordance with the manufacturer's instructions for calibration (see 3.11 and 4.4.3).

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each water bath and replacement pan shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

5.1.1.3 Unit package.

5.1.1.3.1 Type I. Each water bath shall be enclosed in a 3 mil polyethylene bag and adequately cushioned within a box conforming to PPP-B-636, class domestic. Closure of the box shall be as specified in the box specification.

5.1.1.3.2 Type II. Each replacement pan shall be packaged in accordance with MIL-P-116, method III.

5.1.2 Level C. The level C preservation for type I water baths or type II replacement pans shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Four unit packages containing type I water baths or six unit packages containing type II replacement pans shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Four unit packages containing type I water baths or six unit packages containing type II replacement pans shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for type I water baths or type II replacement pans shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Type I water baths or type II replacement pans packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Types I water baths or type II replacement pans packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil agencies. Unless otherwise specified, markings for shipment shall be as specified in FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Level A, B, and C. Each unit pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Number and date of this specification.
(b) Type required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:49Z
Check the source to verify that this is the currentversion before use.

6.2 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

NSN	ITEM IDENTIFICATION
6520-00-535-7100	BATH, WATER, DENTAL IMPRESSION MATERIAL, Thermostatically Controlled, 90°F to Boiling, 115 volt, 50/60 Hz, AC.

6.3 Subject term (key word) listing:
Bath
Dental Impression Compound
Thermostatic control
Water Bath

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS

48661

Preparing activity:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1925

Location: NCR Disk #3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
Attn: DPSC-RSTS
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-B-1400	BATH, WATER, DENTAL IMPRESSION COMPOUND

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
Check the source to verify that this is the current version before use.